Hung Dac Nguyen

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EDUCATION

MICHIGAN STATE UNIVERSITY, College of Natural Science **East Lansing, MI, USA**
Bachelor of Science in Astrophysics, Computational Math, Data Science minor Aug 2023 - May 2027
GPA: 4.0/4.0, Honors College, Dean's List Fall 2023 - Fall 2025

HONORS AND AWARDS

- **Dean's Research Scholar**, College of Natural Science, 2026-2027
- **Hymen and Miriam Stein Endowed Scholarship**, Honors College, Spring 2026
- **College of Natural Science Summer Undergraduate Research Award**, Summer 2025 &2026
- **Cornerstone Undergraduate Research Award**, Department of Earth and Environmental Sciences, 2024 - 2026
- **CMSE Undergraduate Research Award**, Spring 2026
- **Department of Earth and Environmental Science Travel Grant**, Spring 2025 & 2026
- **Honors College Travel Grant**, Honors College, Spring 2025
- **Professorial Assistantship**, Honors College, 2023 - 2025
- **Honors College Excellence Scholarship**, Honors College, 2023 - 2027

RESEARCH EXPERIENCE

SUMMER RESEARCH INTERN - Dr. Kiyoaki Doi and Dr. Haochang Jiang **Heidelberg, Germany**
Max Planck Institute for Astronomy June 2026 - August 2026

- Created a self-consistent thick disk model to solve for the dust and temperature profiles of a protoplanetary disk by numerically solving differential equations and leveraging the radiative transfer code `RADMC-3D`
- Utilized ALMA data and simulation results to reproduce the dichotomy between millimeter and infrared observations for HD 163296

UNDERGRADUATE RESEARCHER - Prof. Seth Jacobsons's Lab Group **East Lansing, MI**
Michigan State University Sep 2023 - Present

- Modified N-body Differentiation program code in C++ to simulate geochemical evolution during lunar accretion and moonlet impacts; utilized GitHub to update code with graduate student mentor and PI
- Leveraged computing clusters to run N-body Differentiation simulations using Bash, Fortran, C++, and used Python to analyze 988 suites of simulations of early solar system formation and lunar accretion; results on lunar formation will be published on Icarus

UNDERGRADUATE RESEARCHER - Prof. Adina Feinstein's Lab Group **East Lansing, MI**
Michigan State University June 2025 - Present

- Reduced spectroscopy and exoplanet transits for 10 targets using Python to identify chemical signatures in exoplanet atmospheres and determine radial velocity trends
- Drafted a JWST proposal on mid-infrared observations of protoplanetary disks around Very Low Mass Stars as probes for terrestrial-planet forming regions, to be submitted in the next cycle

MORP OBSERVER - Prof. Joey Rodriguez & Prof. Laura Chomiuk **East Lansing, MI**
Michigan State University June 2025 - Present

- Active member of the MSU Observatory Research Program; Operated the MSU Observatory's 24-inch reflector telescope using NINA control software to take light curves of exoplanet transits
- Reduced terabytes of nightly observations using AstroImageJ and Python to identify exoplanet transits; combined data with past observations and used AstroPy, Exoplanet, and PyMC machine learning models to fit light curves and extract key orbit parameters
- Instigated interactions with the local community by hosting public outreach nights, setting up telescope arrays to observe the Moon and planets; gave public lectures on general astronomy and solar system processes

PROFESSIONAL EXPERIENCE

UNDERGRADUATE LEARNING ASSISTANT **East Lansing, MI**
ISP 205 - Visions of the Universe Jan 2026 - Present

- Selected by the Department of Physics and Astronomy at Michigan State University to be the undergraduate learning assistant for a sophomore-level natural science course on the conception of Astronomy
- Appointment includes responsibilities in grading, student support, and instructional assistance

RESIDENT ASSISTANT **East Lansing, MI**
Michigan State University Aug 2025 - Present

- Facilitated a multicultural living space for 96 residents by hosting events that promote relaxation and connection; addressed emergencies with punctuality; maintained effective communication up the chain, contributing to a 50% increase in satisfaction

DATA/ AI ENGINEER INTERN **Hanoi, Vietnam**
Kyanon Digital Jun 2024 - Aug 2024

- Created an anomaly detection algorithm in Python, incorporating statistics, machine learning, and regression models to resolve customer exploitation of loyalty program services; presented findings to Department of AI Research

CULTURAL AMBASSADOR **Hanoi, Vietnam**
19 Nguyen Sieu Jul 2024 - Jan 2025

- Hosted international visitors at the historic 19 Nguyen Sieu cultural heritage site; facilitated seamless communication between English, Vietnamese, and Chinese, ensuring an inclusive experience for tourists

PUBLICATIONS AND CONFERENCE PRESENTATIONS

- **Hung D. Nguyen**, Gabriel Nathan, Seth A. Jacobson (In preparation). Lunar core and mantle abundances can be reproduced from a bulk-silicate Earth composition when starting with Theia core.
- **Hung D. Nguyen**, Gabriel Nathan, Seth A. Jacobson. *Lunar core and mantle abundances can be reproduced from a bulk-silicate Earth composition when starting with Theia core.* Poster presented at the 57th Division of Dynamical Astronomy Annual Meeting (DDA), Chicago, IL, June 2026
- **Hung D. Nguyen**, Gabriel Nathan, and Seth A. Jacobson. *"Oxidation state gradient in the protoplanetary disk: piecewise or smooth."* Poster presented at Lunar and Planetary Science Conference 2025 (LPSC), The Woodlands, TX, March 2025.
- **Hung D. Nguyen**, Gabriel Nathan, and Seth A. Jacobson. *"Theia core and mantle contributions to the protolunar disk [abstract]."* Poster presented at Michigan Space Grant Consortium 2025 Fall Conference (MGSG), Central Michigan University, MI, October 2025.
- **Hung D. Nguyen**, Gabriel Nathan, and Seth A. Jacobson. *"Baking the Moon: Reproducing the Earth and Moon System Using Numerical Simulations."* Poster presented at the Mid-Michigan Symposium for Undergraduate Research Experiences (MidSURE), East Lansing, MI, July 2025.
- **Hung D. Nguyen**, Gabriel Nathan, and Seth A. Jacobson. *"Baking" the Solar System: Investigating the Chemistry of Terrestrial Planet Formation."* Poster presented at the 2025 University Undergraduate Research and Arts Forum (UURAF), East Lansing, MI, April 2025.
- Seth A. Jacobson, Gabriel Nathan, **Hung D. Nguyen**. *"Theia can arrive late and be oxidized, but not if it is large compared to proto-Earth."* Poster presented by Seth A. Jacobson at the Europlanet Science Congress - Division of Planetary Science Conference 2025 (EPSC-DPS), Helsinki, Finland, September 2025.
- Gabriel Nathan, **Hung D. Nguyen**, and Seth A. Jacobson. *"Solar system formation scenarios effect on ideal redox state of disk needed to produce Earth."* Oral presentation by Gabriel Nathan at the Goldschmidt Conference, Chicago, IL, August 2024.

SKILLS

Programming & Analysis: Python, R, Bash, C++, Fortran
Scientific Tools: RADMC-3D, Athena++, CARTA, NINA, AstroImageJ, SLURM, Git, LaTeX, GitHub, SAOImageDS9
Languages: English, Vietnamese, Chinese
Technical Tools: Adobe Premiere Pro

REFERENCES

Dr. Seth A. Jacobson, Ph.D., *current supervisor* seth@msu.edu
Assistant Professor, Department of Earth and Environmental Sciences, Michigan State University

Dr. Adina Feinstein, Ph.D., *Exoplanet research advisor* adina@msu.edu
Assistant Professor, Department of Physics and Astronomy, Michigan State University

Dr. Joey Rodriguez, Ph.D., *MORP supervisor* jrod@msu.edu
Associate Professor, Department of Physics and Astronomy, Michigan State University